

Play

Iteration 2

⊗ Failed



Proposed Task

"I want to see if I can get the drawer on this sidetable by pulling its handle."

Policy / Low-level Plan (Attempt)

```
obs = get_observation()
handle_pos = localize_object("drawer handle")
grasp_pose = get_horizontal_grasp(handle_pos)
open_gripper()
pos, quat = decompose_transform(grasp_pose)
goto_pose(pos, quat)
close_gripper()
approach_axis = grasp_pose[:3, 2]
pull_pos = pos - 0.15 * approach_axis
goto_pose(pull_pos, quat)
# pull along a wrong axis
open_gripper()
```

failed to open drawer

Invoked under failed attempts

Skill Proposer



Reviews previous failures and proposes reusable helpers.

get_axis_aligned_pull_direction(target_pos)

select_grasp_for_pulling(grasps, scores, pull_dir)

Play

Iteration 16

○ Success



Proposed Task

"I'm curious if I can pull the drawer handle towards me."

Policy / Low-level Plan (Uses Skills)

```
def plan_pull_grasp(object_name: str):
    target_pos = localize_object(object_name)
    pull_dir = get_axis_aligned_pull_direction(target_pos)

    obs = get_observation()
    grasps, scores = plan_grasp(obs)
    best_grasp, best_score = select_grasp_for_pulling(grasps, scores, pull_dir)
    return best_grasp, pull_dir

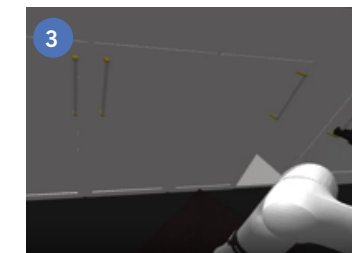
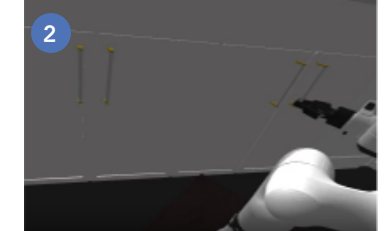
grasp_pose, pull_dir = plan_pull_grasp("drawer handle on the desk")
pos, quat = decompose_transform(grasp_pose)
goto_pose(pos, quat)
close_gripper()
pull_pos = pos - 0.15 * pull_dir
goto_pose(pull_pos, quat)
open_gripper()
# drawer opened
```

Evaluation using Skills

Evaluation Task

"Open the Cabinet"

○ Success



Policy (Evaluation Rollout using Skills)

```
obs = get_observation()
mask = get_handle_mask(obs)
target = localize_object(mask)
pull_dir = get_axis_aligned_pull_direction(target)
pts_world, pc_full = get_point_clouds_from_mask(mask)
grasps, scores = plan_grasp(pts_world, label="cabinet handle")
grasp_pose, score = select_grasp_for_pulling(grasps, scores, pull_dir)

pos, quat = decompose_transform(grasp_pose)
approach_axis = grasp_pose[:3, 2]
move_to_offset(pos, approach_axis)
grasp_and_pull(pos, quat, pull_dir)
open_gripper()
# drawer opened
```

Early attempt to pull the sidetable drawer handle fails.

Agent analyzes failures and proposes reusable helpers.

Later attempt reuses skills and successfully opens drawer.

Skills generalize to a new task and complete the evaluation successfully.